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ADMISSION NUMBER: BSCCS/2024/50482

COURSE/CLASS: COMPUTER SCIENCE 2024 SEP CLASS

UNIT CODE: BIT3208

UNIT: ADVANCED WEB DESIGN AND DEVELOPMENT

LECTURE NAME: MICHAEL NYORO

PROJECT TITLE: Apex Library Management System.

TECHNOLOGIES USED:

- **Backend Engine:** PHP for session validation, and circulation calculations.
- **Database Engine:** MySQL.

GITHUB LINK: <https://github.com/t4g1z>

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WEEK ONE: Xampp Local Environment Setup

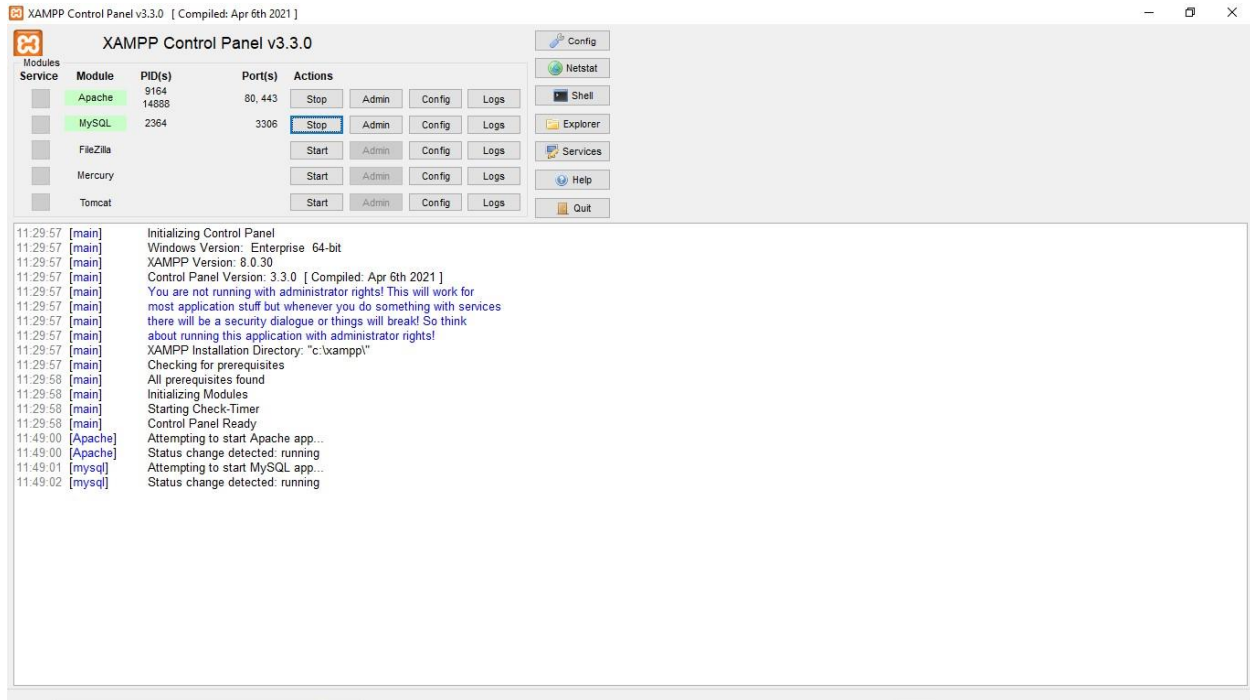


Fig 1.1 Apache and MySQL Running

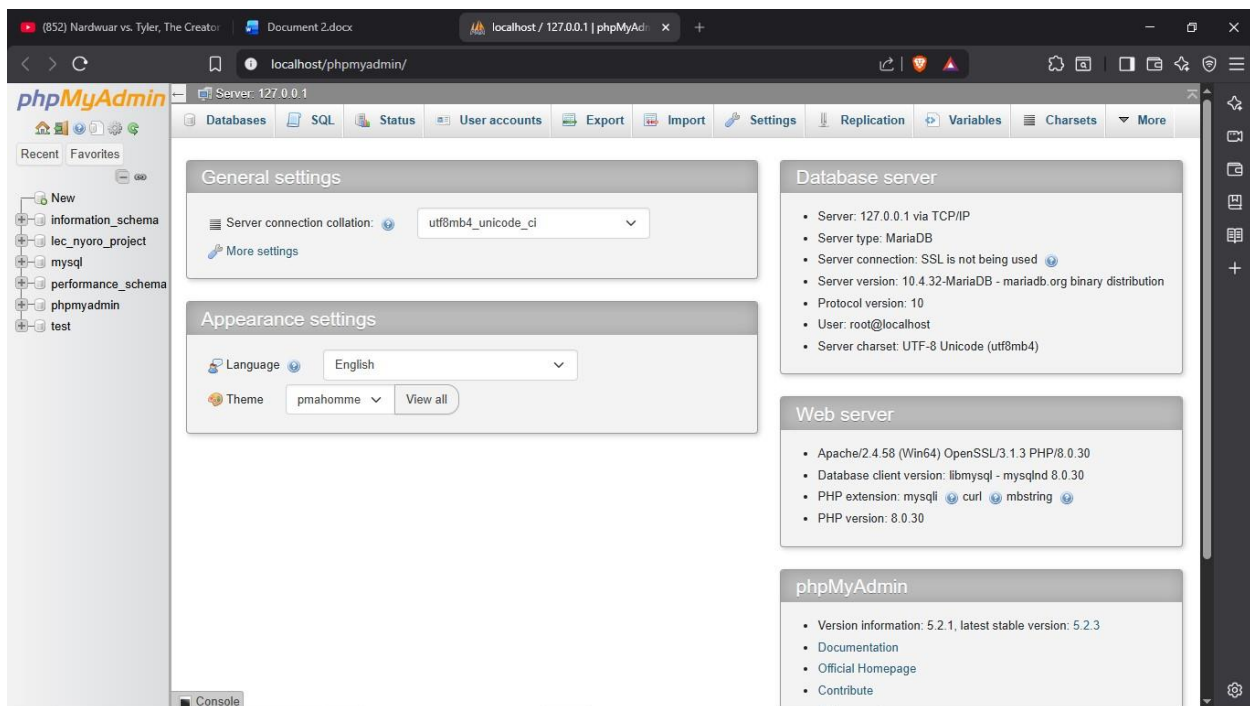


Fig 1.2 Local host test page

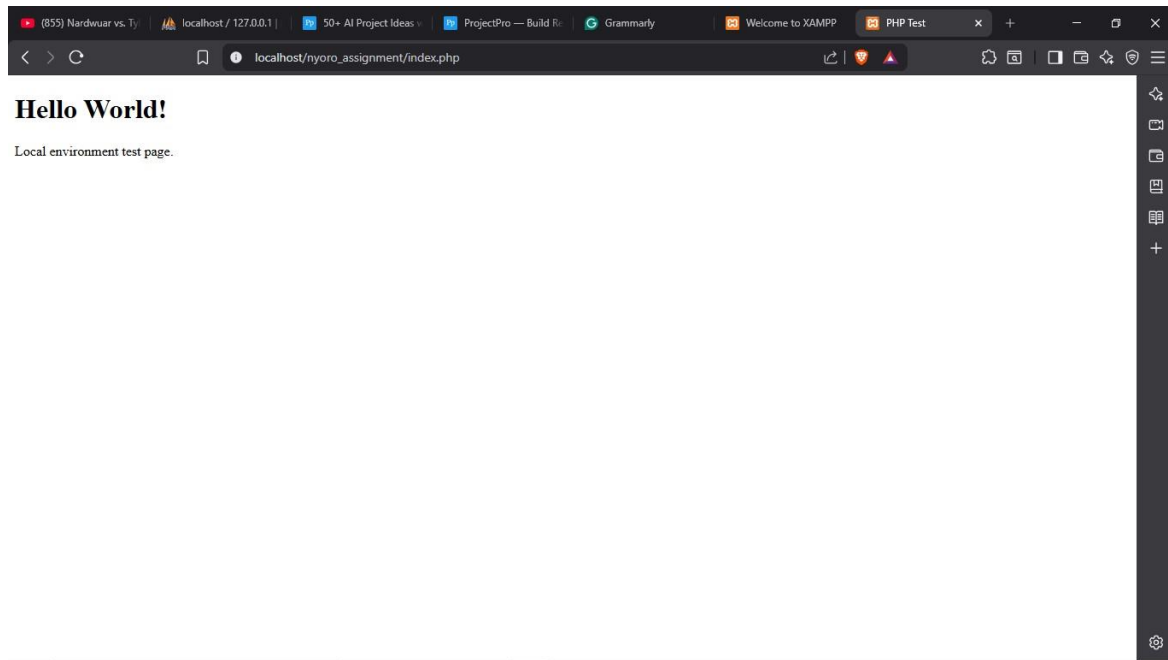


Fig 1.3 Mock page displaying HELLO WORLD

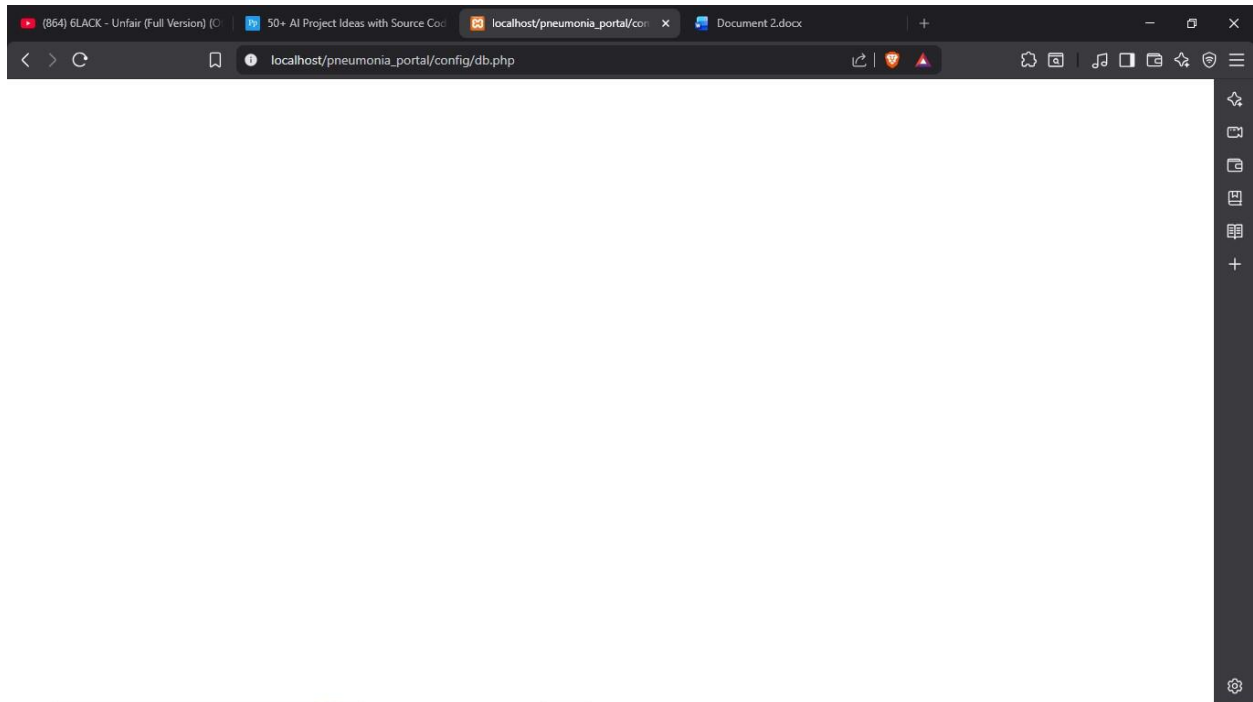


Fig 1.4 Shows a blank page to confirm that the connection to MySQL has no errors and is connected successfully.

What I learnt

Mastered local server architecture deployment by configuring XAMPP (Apache and MySQL). Learned the difference between client-side and serverside execution and organized a modular, professional project directory structure for proper software development workflow.

WEEK 2: UI/UX AND DESIGN FOUNDATIONS

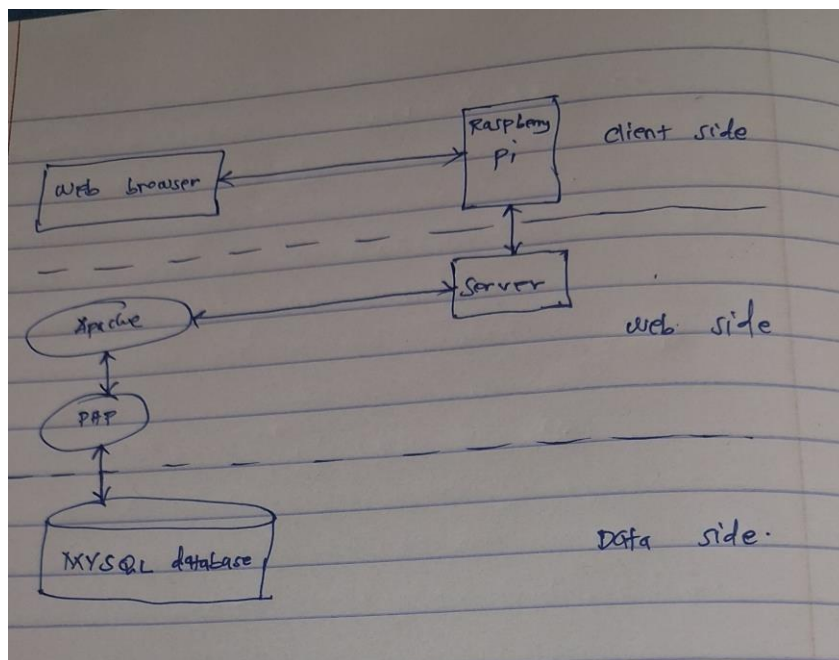
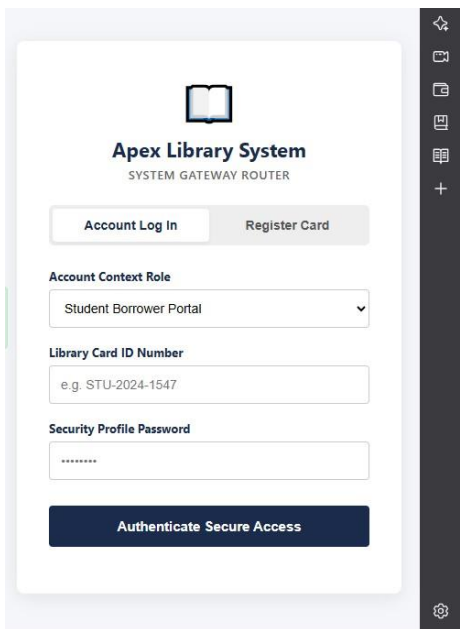
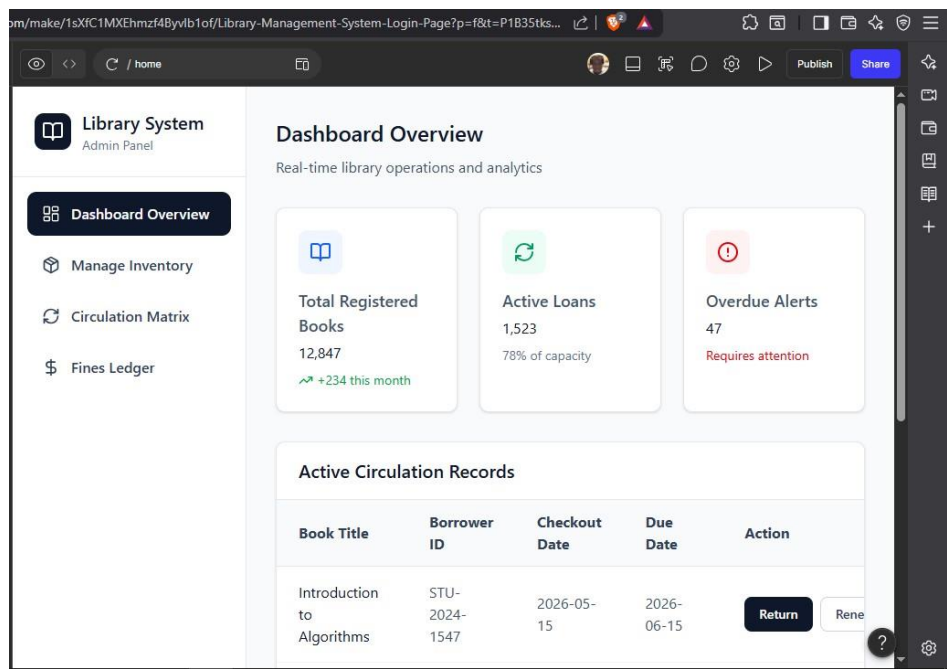


Fig 2.1 Client-Server Data Distribution Architecture Model.



The image shows a login/register form for the Apex Library System. The form is titled "Apex Library System" with the subtitle "SYSTEM GATEWAY ROUTER". It features two buttons: "Account Log In" and "Register Card". Below these, there is a dropdown menu for "Account Context Role" with "Student Borrower Portal" selected. A text input field for "Library Card ID Number" contains the example "e.g. STU-2024-1547". A password input field for "Security Profile Password" is shown with masked characters. A large blue button at the bottom is labeled "Authenticate Secure Access". The form is set against a light purple background with a dark sidebar on the right containing various icons.

Fig 2.2 Figma wireframe to show the login/registration form.



The image is a Figma wireframe of a "Library System Admin Panel" dashboard. The left sidebar contains navigation links: "Dashboard Overview" (selected), "Manage Inventory", "Circulation Matrix", and "Fines Ledger". The main content area is titled "Dashboard Overview" with the subtitle "Real-time library operations and analytics". It features three summary cards: "Total Registered Books" (12,847, +234 this month), "Active Loans" (1,523, 78% of capacity), and "Overdue Alerts" (47, Requires attention). Below these is a section for "Active Circulation Records" containing a table with columns: Book Title, Borrower ID, Checkout Date, Due Date, and Action.

Book Title	Borrower ID	Checkout Date	Due Date	Action
Introduction to Algorithms	STU-2024-1547	2026-05-15	2026-06-15	<button>Return</button> <button>Renew</button>

Fig 2.3 Figma wireframe to show dashboard.

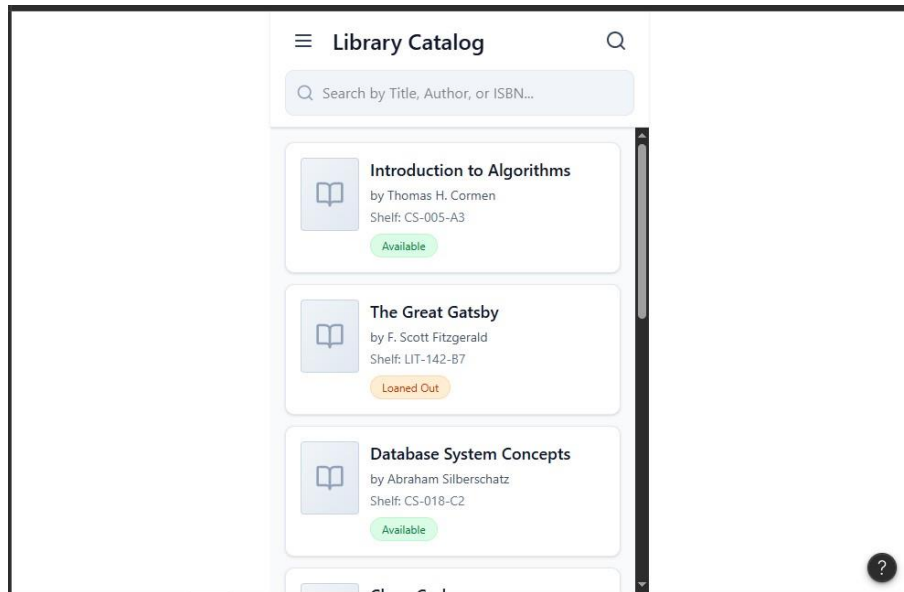


Fig 2.4

Fig 2.2, 2.3 and 2.4 show Figma wireframe designed for the intended system.

What I learnt

Explored design thinking and user experience paradigms by creating lowfidelity UI/UX structural wireframes in Figma. Developed responsive interfaces for desktop and mobile form screens, utilizing standard layout placeholder notations to map out the application's visual flow.

WEEK 3: FRONTEND SCRIPTING AND BACKEND PROGRAMMING FOUNDATIONS

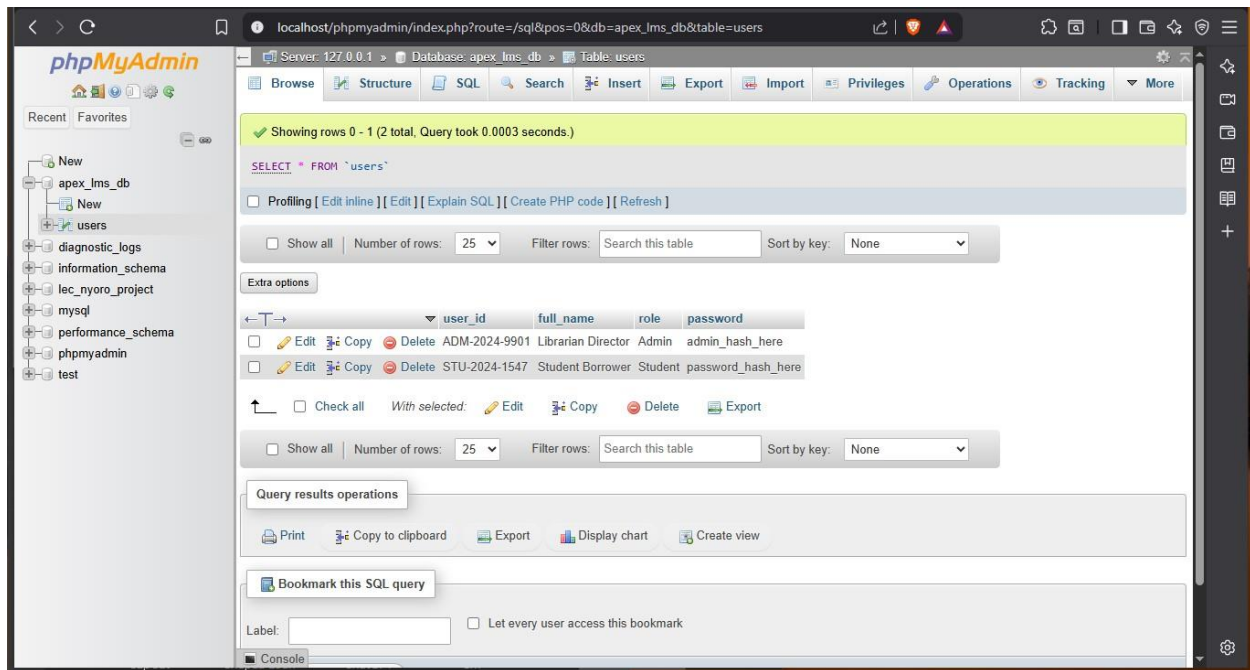


Fig 3.1 Shows my database for apex library management system.

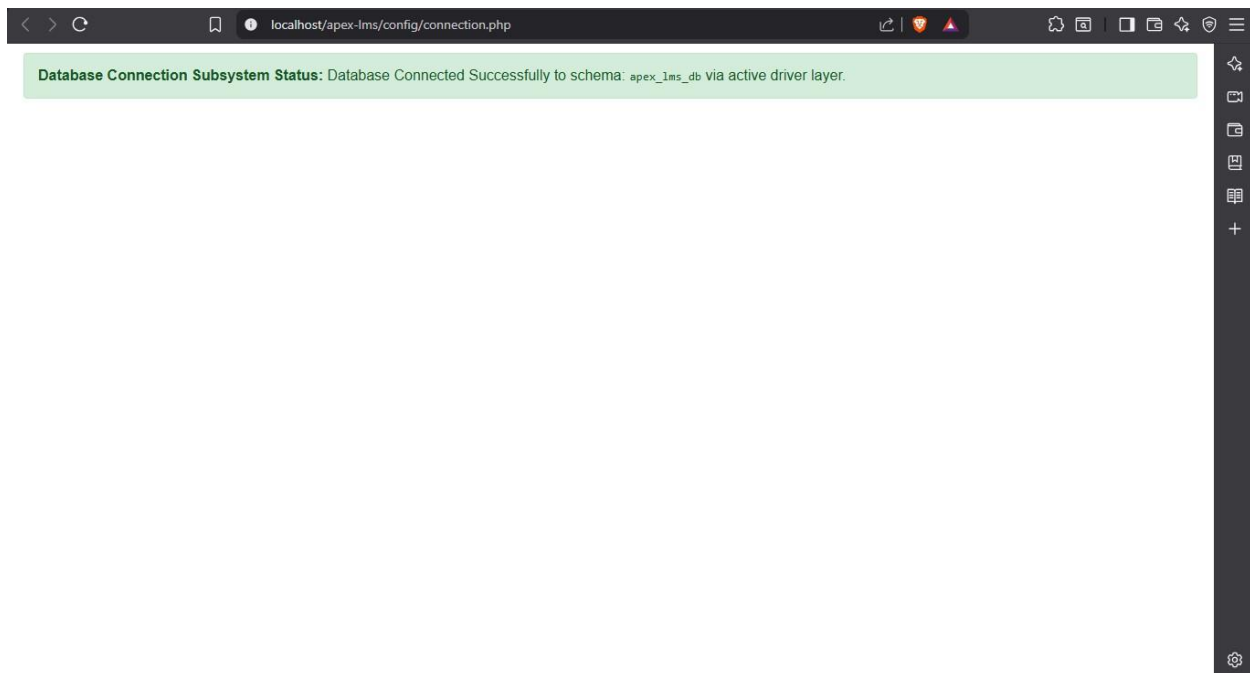


Fig 3.2 Shows that the database is connected successfully.

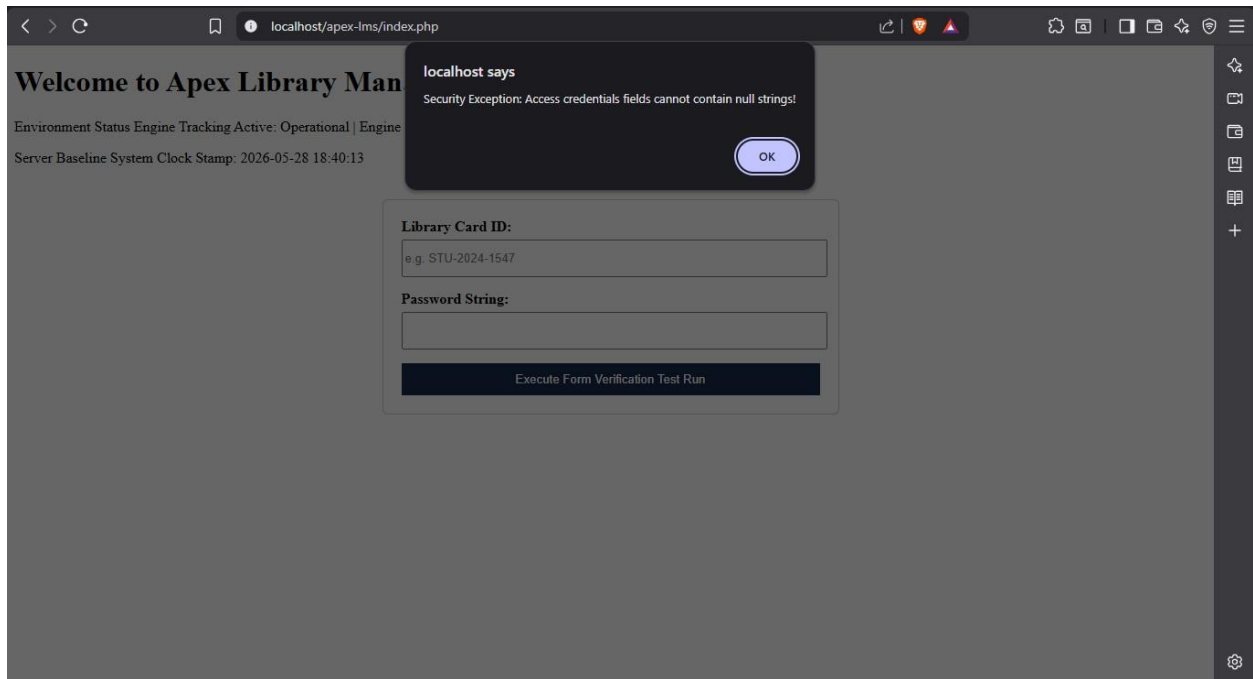


Fig 3.3 Shows error when the login page has nothing inserted in the fields.

What I learnt

Introduced client-side scripting and backend foundations. Built a form validation system using JavaScript DOM event manipulation, implemented a real-time password strength analyzer, and established a secure, objectoriented database connection bridge using modern PHP Data Objects (PDO) strings.

GITHUB VERSION CONTROL 1:<https://github.com/t4g1z/apex-lib>

WEEK 4: DYNAMIC BACKEND FORM PROCESSING & SYSTEM AUTHENTICATION

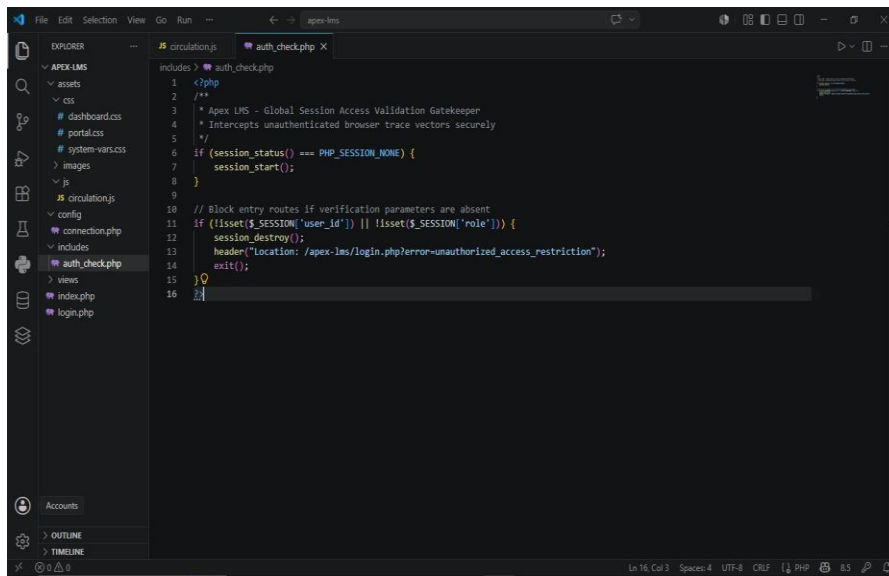


fig 4.1 Security Gatekeeper Middleware.

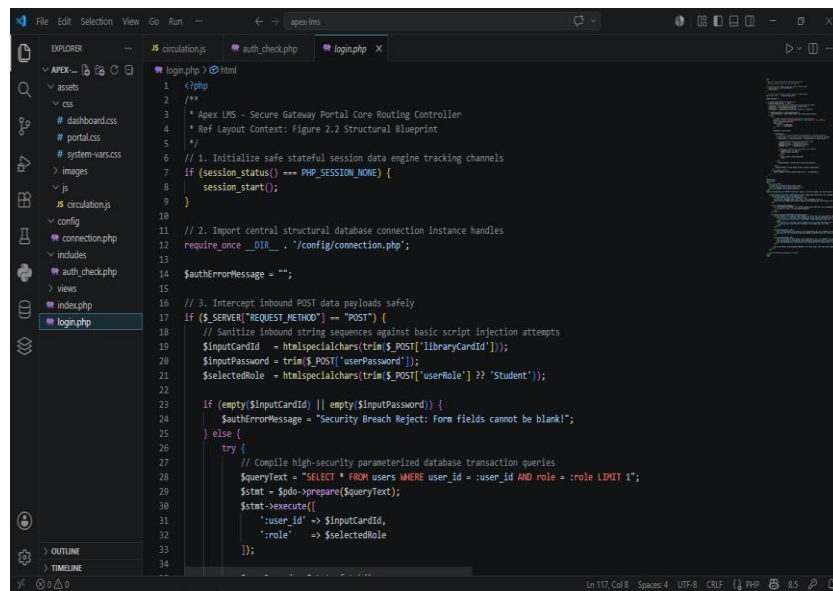


Fig 4.2 Login page form code.

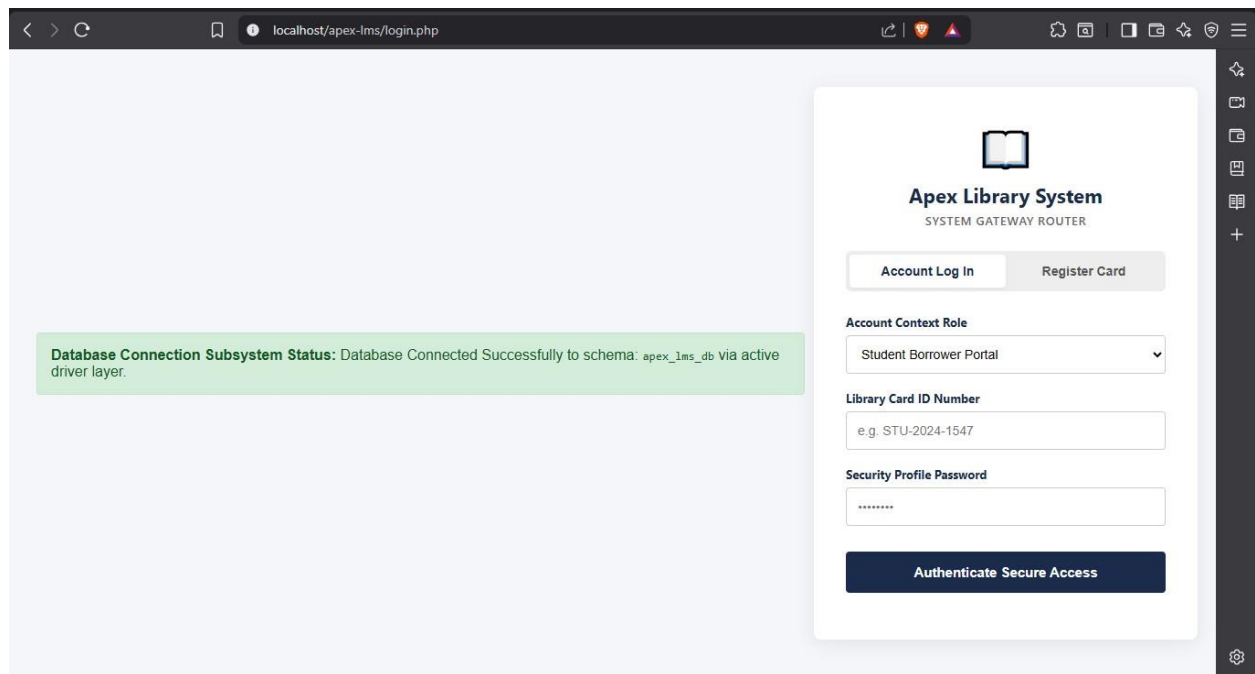


Fig 4.3 Login page active.

What I learned

Learned the full HTTP request-response cycle by integrating HTML forms with backend processing. Implemented secure data handling via the POST method, introduced user authentication pipelines using session tracking (`$_SESSION`), and created a defensive middleware gatekeeper file.

WEEK 5: DATABASE COMPLETION AND DYNAMIC CRUD SYSTEM LAYER INTEGRATION

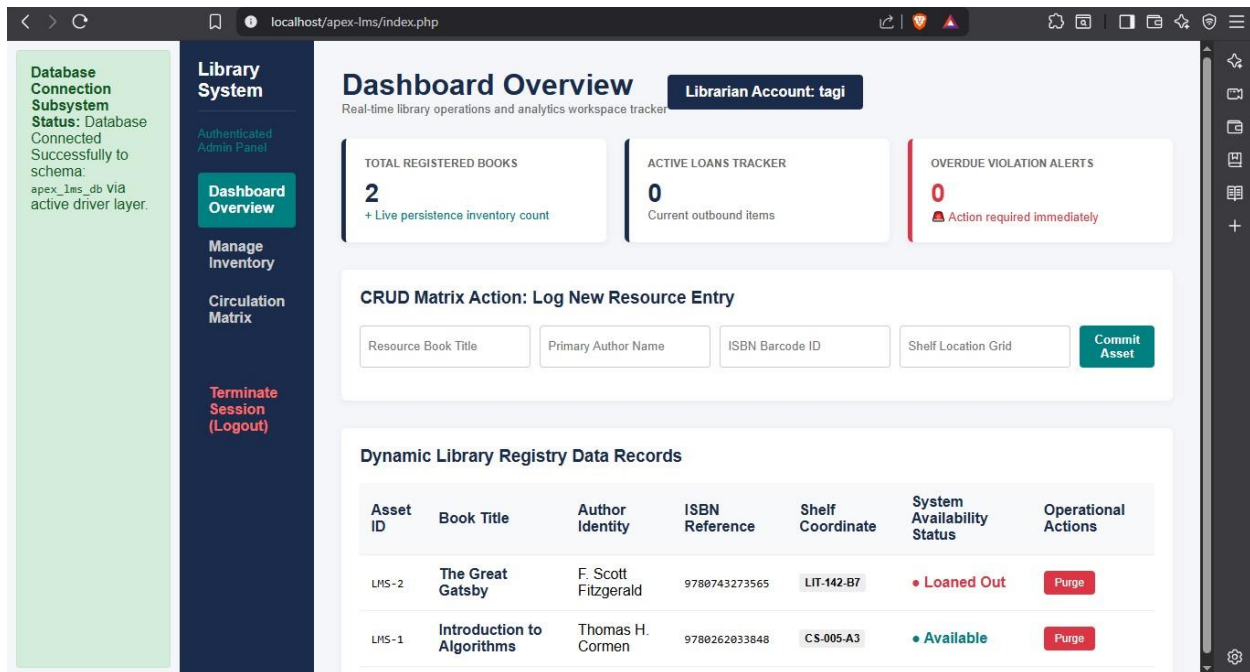


Fig 5.1 Shows the final dashboard page after login

What I learned

Implemented full backend data-driven integration. Constructed relational MySQL schemas with foreign key constraints, mastered full CRUD mechanics (Create, Read, Update, Delete) to dynamically add, display, and purge database assets, and effectively debugged system runtime errors like SQLSTATE base table exceptions.

GITHUB VERSION CONTROL 2: <https://github.com/t4g1z/apex-lib/tree/main/apex-lms>

WEEK 6: Implementation of Multi-Module CRUD Systems & Relational Database Integration

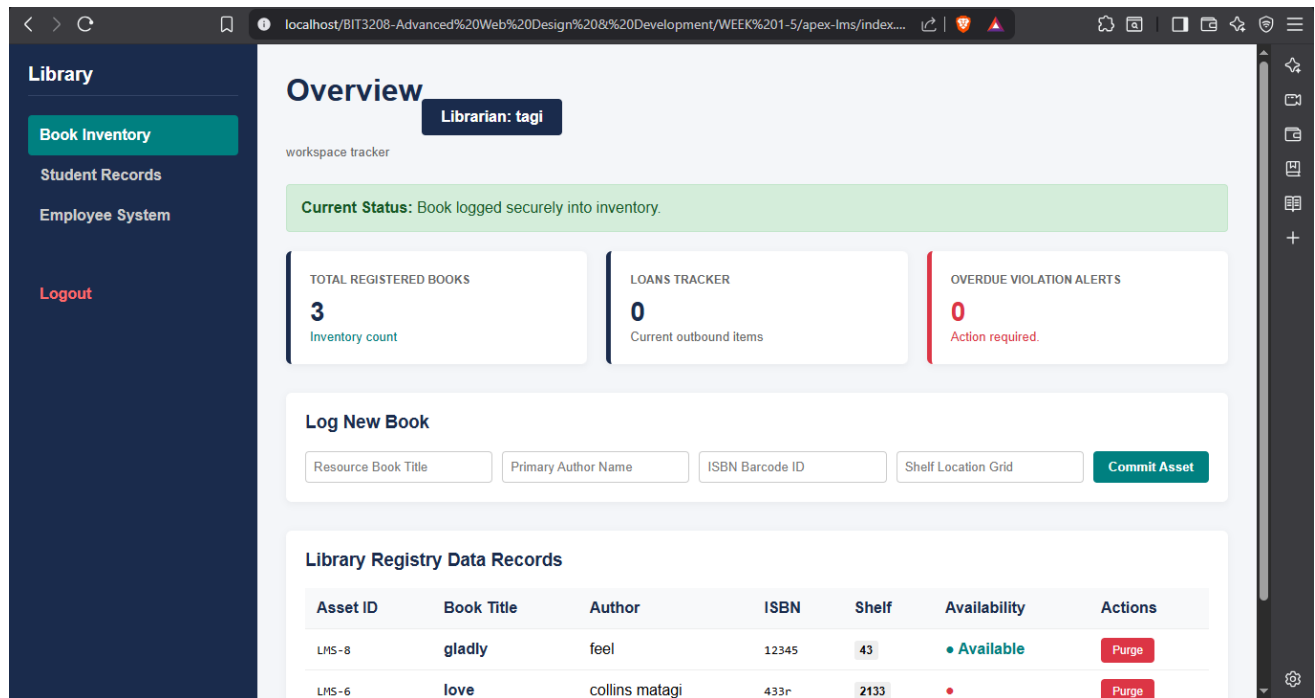


Fig 6.1 Shows successful creation of a book called gladly and inserted into the database.

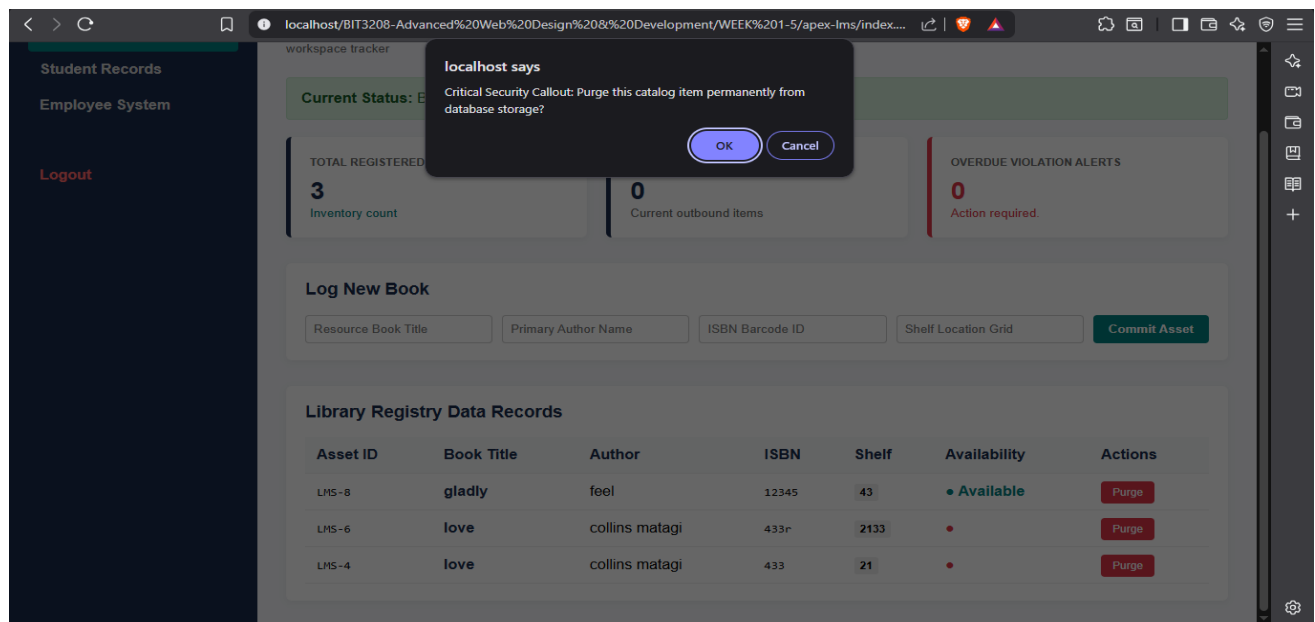


Fig 6.2 Shows successful deletion of a book.

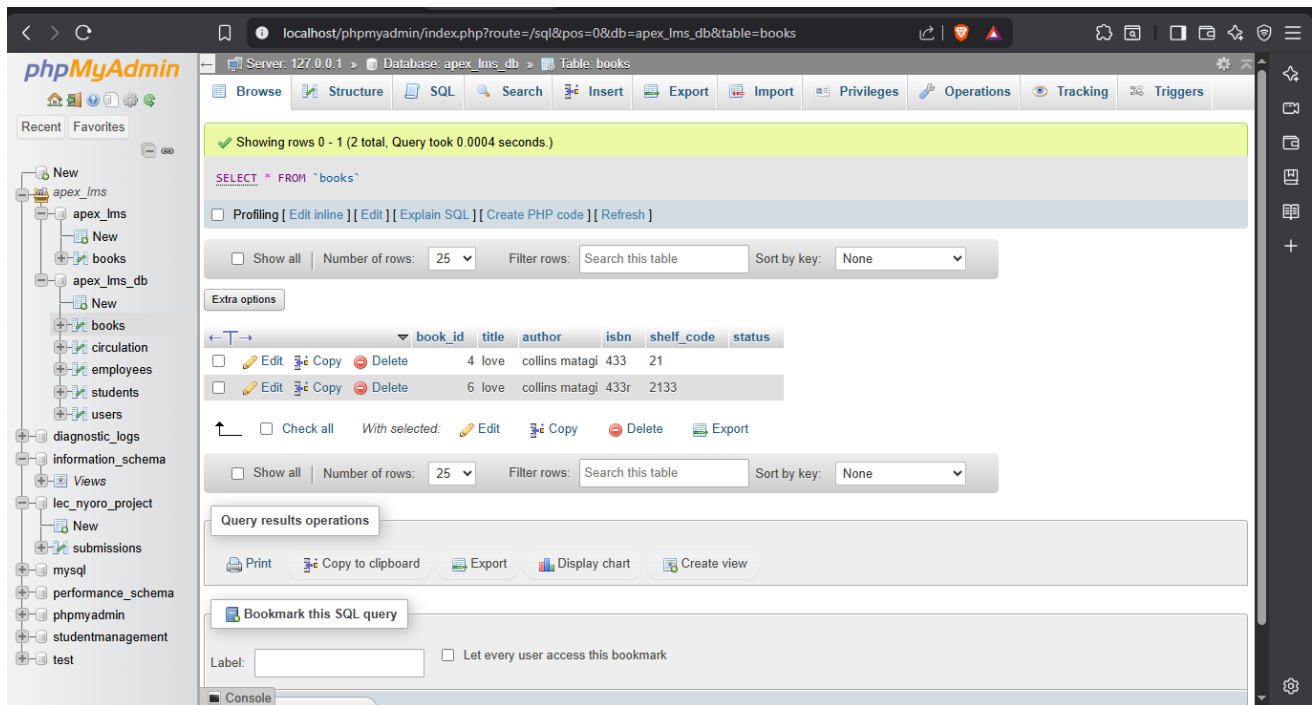


Fig 6.3 shows that the books added reflect in the database.

What I learned

I got to clearly understand different operations in database manipulation on my local MySQL server.

Week 7: Role-Based Access Control (RBAC) Matrix, Multi-User Authentication Gateways, and Stateful Session Management

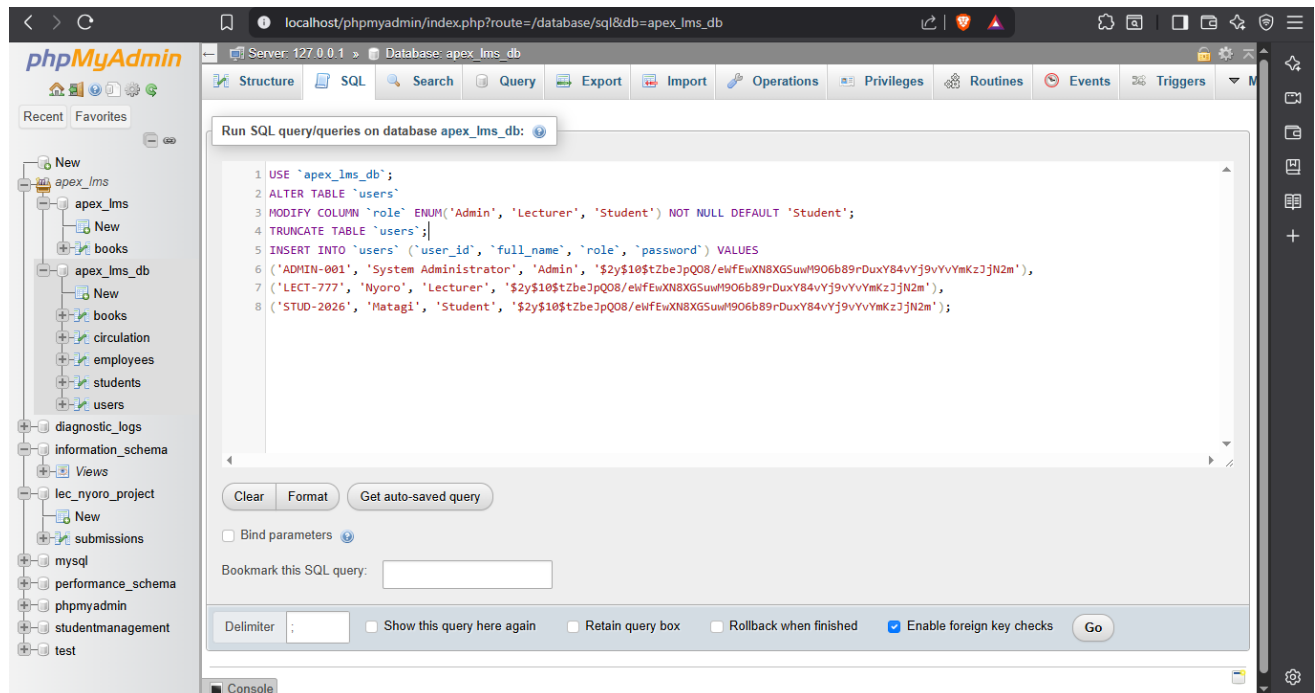


Fig 7.1 Shows advancing the system by altering the user's datatype to introduce privileges.

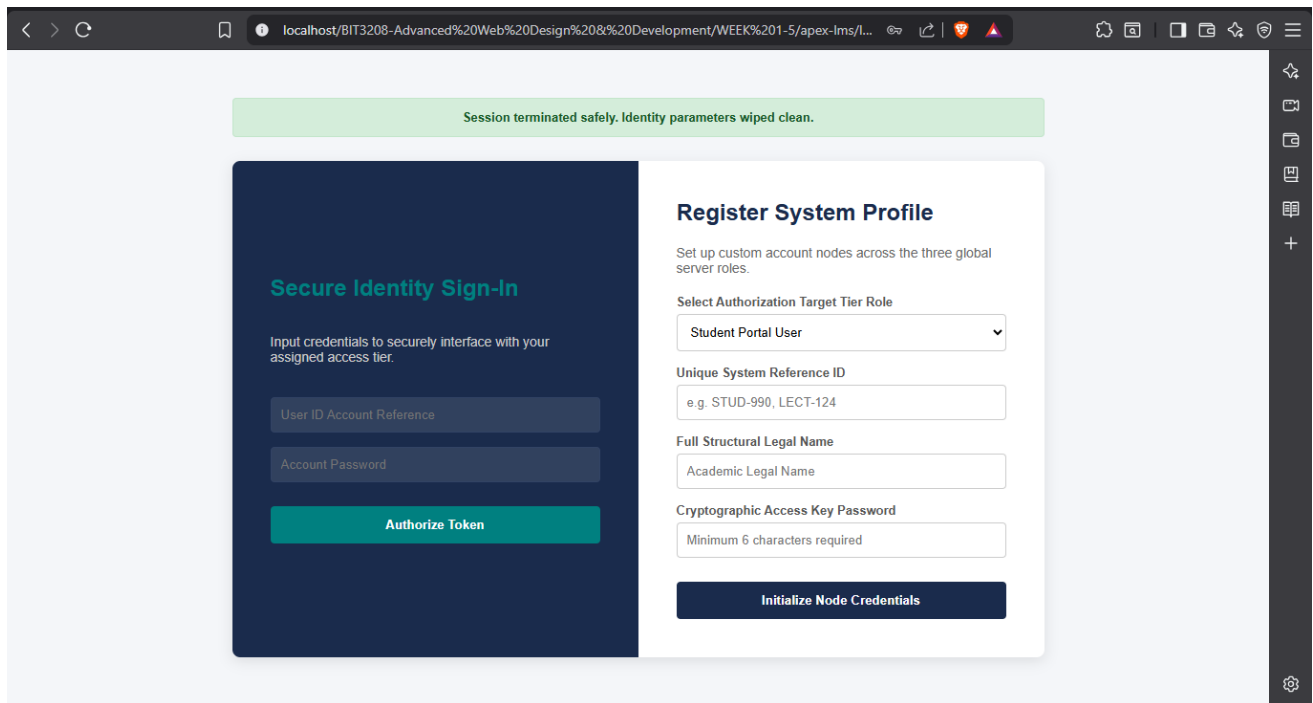


Fig7.2 shows the new login page after updating the student and admin roles and access.

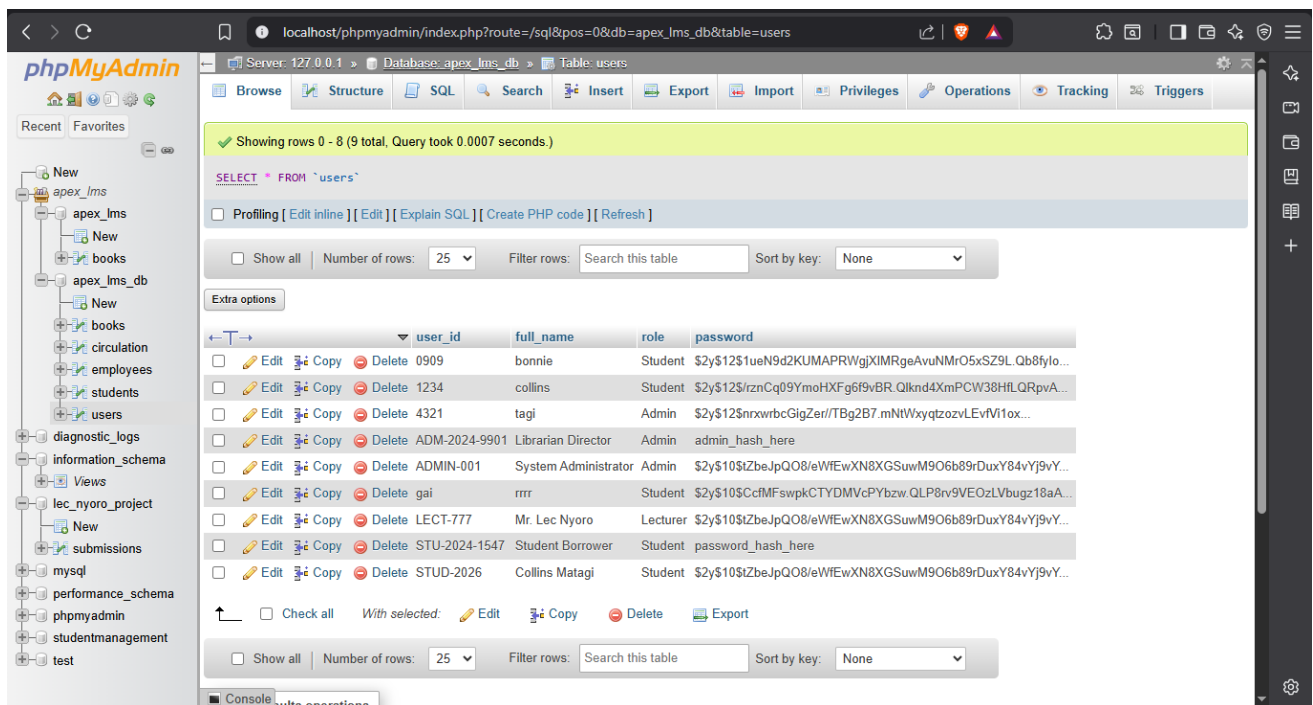


Fig 7.3 shows the roles of each user clearly in the database.

What I learned

Gained a solid understanding of structural Role-Based Access Control configurations.